

Supplementary Information for

When do single-cell response distributions improve drug retrieval?

This Supplementary Information provides additional methodological detail, robustness analyses and supporting experiments for the main text. It contains Supplementary Methods, four Supplementary Notes and four Supplementary Tables, and no Extended Data figures. The supplementary analyses follow the same conceptual structure as the main text: they examine the information retained by population-level retrieval, the dependence of apparent retrieval gains on the evaluation criterion, and the conditions under which state-specific single-cell responses can become consequential for candidate ranking.

11 Supplementary Methods

12 **Implementation of population retrieval scores.** Energy distance was the primary population score.
13 RBF-kernel maximum mean discrepancy used the median squared pairwise distance of the pooled
14 query and candidate sample as its bandwidth reference and evaluated scales of 0.25, 1 and 4 times
15 that value. Sliced Wasserstein-1 distance averaged quantile-matched projections over 128 fixed ran-
16 dom directions. Candidate populations were subsampled with a fixed seed to at most 500 cells in the
17 observed-population analyses and 300 cells in the partial-observation analyses. A population containing
18 fewer than two cells after filtering was assigned a fixed large penalty distance rather than removed from
19 the candidate library.

20 **Implementation of subpopulation coverage.** Query cells were partitioned by k -means with $k = 2$,
21 three initializations and a fixed seed. Candidate cells were assigned to the query component whose
22 centroid had the highest cosine similarity. A candidate with no cell assigned to a query component
23 received the penalty distance for that component. At small β , the coverage distance was evaluated using
24 $D_\beta = \text{mean}(d) + \frac{\beta}{2} \text{var}(d) + O(\beta^2)$ to avoid numerical cancellation in the log-sum-exp form. Coverage-
25 mean and coverage-worst use the $\beta \rightarrow 0$ and $\beta \rightarrow \infty$ limits, respectively. The base distance was energy
26 distance unless otherwise specified; the complete score registry is given in Supplementary Table 1.

27 Supplementary Note 1. Additional diagnostics of when population information changes 28 ranking

29 The main text separates the existence of state-specific perturbation responses from their recoverability
30 and from the downstream requirement that they alter candidate ordering. We performed several ad-
31 ditional analyses to ask whether simple observables could identify queries for which population-level
32 retrieval is more useful.

33 **Query identifiability.** In A549 leave-drug-out retrieval, query identifiability was quantified as a
34 two-cluster silhouette on the query cells, independently of the retrieval outcome. Across 543 evalu-
35 able query-seed combinations it was weakly and non-monotonically associated with the absolute
36 mechanism-of-action nDCG of a single energy-distance run (Spearman $\rho = 0.17$, $p = 6.3 \times 10^{-5}$), and
37 the association is confounded with candidate-library composition: the highest-identifiability tertile
38 holds a median of six same-mechanism candidates against two in each lower tertile, and a larger
39 relevant pool mechanically raises the attainable nDCG. The median silhouette score even in that tertile
40 is 0.038, so the two-state structure these queries carry is weakly resolved. This analysis scores one
41 retriever and therefore bears on absolute recovery rather than on the population-minus-mean gain.

42 **Unsupervised recovery of constructed two-state structure.** Response-identity recoverability in the
43 constructed two-state mixture is a supervised ceiling against the best unsupervised method (ceiling 0.692
44 against best unsupervised 0.674), and separately a ceiling of 0.879 when the same cells are posed at the
45 level of individual drug pairs rather than pooled drug classes. Three further measurements characterise
46 the same limit.

47 First, across nine unsupervised clustering configurations, the median adjusted Rand index against
48 the known two-source labels ranged from 0.037 to 0.106. The configurations were k -means with $k = 2$
49 in the raw, 10-dimensional PCA, 50-dimensional PCA and response spaces, two-component Gaussian
50 mixtures in the raw and 50-dimensional PCA spaces, and k -means with k selected from 3, 4 and 5 by
51 silhouette, reported separately by which k was selected. The highest value, 0.106, was obtained when
52 silhouette selection chose $k = 4$. Recovery is weak under all nine, so the low ceiling is not specific to the
53 clustering method used in the main analysis.

54 Second, the weak separation is visible in the raw geometry rather than only in an accuracy score. Sil-
55houette scores computed independently for each perturbation-response population under a two-cluster
56 partition had a median of 0.034 in SciPlex3 ($n = 529$ populations) and 0.084 in Frangieh ($n = 231$);
57 in both datasets fewer than 0.5% of populations exceeded 0.1. Visible multimodality is therefore not a
58 prerequisite for, and should not be equated with, recoverable state-specific perturbation information.

Third, recovery is limited by state separation rather than by cell number. Averaging the k -means $k = 2$ adjusted Rand index over 20 seeds across a grid of source separation and cells per source, increasing the cell budget from 25 to 400 cells per source changed recovery by at most 0.02 at fixed separation, whereas increasing separation from 1.0 to 8.0 times the observed value raised it from 0.07 to 1.00. The separation transform was defined from the full source populations before subsampling. Additional cells therefore do not compensate for states that are close together, which is the same conclusion the supervised ceiling reaches by a different route.

Measured response divergence. We next stratified the 600 mechanism-evaluable queries by measured response divergence, one minus the mean cosine between their state-specific perturbation responses. Greater divergence was associated weakly with a larger population-minus-mean mechanism-recovery difference (Spearman $\rho = 0.15$, $p = 1.4 \times 10^{-4}$), but the conditional advantage did not become convincingly positive. The most divergent quartile reached a mean nDCG difference of +0.014 with a bootstrap interval of -0.012 to $+0.041$, and the direction reversed in two of the three SciPlex3 cell lines (A549 -0.107 , $n = 9$; K562 -0.021 , $n = 38$; MCF7 $+0.040$, $n = 96$). That quartile is not balanced across cell lines, and its positive mean is carried by the one line that contributes two thirds of it.

These results distinguish differential response from decision relevance. Stronger state-specific responses may create the information required for population-aware retrieval, but they do not guarantee that this information changes the ordering of candidate perturbations.

Precision of the two task-proximal comparisons in the highest-divergence stratum. In the most divergent quartile the two criteria differ in precision rather than in effect size. The mechanism-recovery gap is nearly three times the minority-coverage gap but its per-query standard deviation is twelve times larger (0.169 against 0.0136), so the two readings are asymmetric: the minority-state comparison is well determined and returns a small effect, which is evidence of a small effect, whereas the mechanism-recovery comparison in this stratum supports uncertainty rather than evidence of no effect. Achieved power and the sample size required for 80% power are given in the code release. The main text's mechanism-recovery conclusion rests on the full 600-query set and on the shape of the whole distribution (Fig. 3a,b), not on this stratum.

Instability of thresholded dominance counts. Counting tasks whose population-level gain exceeds a fixed threshold is unstable under resampling: at a gain threshold of +0.01, only 2 of the 10 initially threshold-crossing cross-line tasks stay above it in at least 8 of 10 seed resamples, while control drugs that cross it in no recorded seed cross it in 1 to 4 of 10 resamples. No count of dominant tasks is therefore reported from this work, and the main text reports continuous retrieval differences and stratified analyses instead.

Two checks on the diagnostic's decline verdicts. The pre-specified diagnostic declines 127 queries as "mean or no call" and a further 11 as "mean sufficient". Pooling the 11 into the non-recommended group leaves the verdict unchanged, and every interval involved is wider than either group median, so the comparison is a failure to detect enrichment rather than a demonstration that there is none. The two decline verdicts behave oppositely: the 127 "mean or no call" queries are *more* divergent than the queries the diagnostic recommends (medians 1.68 against 1.66), which is the direction its design does not predict, while the 11 "mean sufficient" queries lie below the fifth percentile of both groups, where the diagnostic is correct.

Supplementary Note 2. Verification and transfer limits of HIR-Bench

HIR-Bench was introduced as a controlled construction in which the conditions under which within-population structure can alter candidate preference are known analytically. It is therefore used to verify the logic of a preference-changing regime rather than to infer a biological phase boundary from data.

104 The generator satisfies its internal consistency checks on the full parameter grid. All three pre-
 105 specified checks pass: when the two subpopulations have no conflict in candidate preference the
 106 retrieval-flip rate is 0 against a threshold of 0.05, as required by construction; the median distance
 107 between the grid’s own mixing fraction and the analytical boundary

$$\alpha^* = \frac{B}{A+B},$$

108 is 0.292, a range guard rather than a discriminating test since the statistic cannot exceed its threshold
 109 of 1.0; and under a predicted-mean information condition the population-level and mean-level Hit@1
 110 differ by 0.051 against a threshold of 0.10, which is the degeneracy the construction requires when no
 111 within-population structure is available to the retriever.

112 The measured counterpart of the analytical boundary behaves as the construction predicts. Over the
 113 36 cells of the conflict-by-majority-fraction grid, the energy distance exceeds mean-signature retrieval in
 114 Hit@1 in 30 of them, and the advantage runs from -0.25 to $+1.00$, largest where conflict is high and the
 115 majority fraction is above α^* and vanishing below it.

116 We next asked whether the synthetic regime could be recognized from quantities observable at re-
 117 trieval time. Observable features included the subpopulation-variance ratio, isotropy index and re-
 118 sponse diversity of the query and candidate populations. Under a cross-validation unit that respects
 119 the 28 label-determining parameter cells, those features reach a pooled out-of-fold AUC of 0.788 against
 120 a majority-class rate of 0.643. Splitting instead on the 672 grid configurations, which leaves about twelve
 121 near-duplicates of each held-out instance in training, raises it to 0.836, so leakage alone is worth $+0.048$
 122 and the honest observable result is modest. Because the label is deterministic within a parameter cell,
 123 every held-out fold is single-class and the pooled out-of-fold AUC is the only statistic the design admits.
 124 Removing within-population information also raised the best decision regret over the method set, which
 125 a predicted-structure condition recovered, so the construction penalizes that loss in the direction it was
 126 built to.

127 The analytical boundary was also projected to the biological retrieval tasks as a transfer test. This
 128 projection did not generalize. The resulting labels were strongly imbalanced (229 versus 10), for which
 129 a constant majority-class predictor achieves 95.8% accuracy. The synthetic-regime projection reached
 130 39.7%, far below this baseline and below it on four of the five datasets separately (0.29, 0.40, 0.33 and
 131 0.22); the fifth is the predicted-mean subset, where the projection returns the same verdict for every
 132 task by construction. The boundary was transferred by percentile rather than by value: the crossover
 133 percentile was fitted on 4,480 grid cells of the HIR-Bench method-performance layer and applied to the
 134 real tasks at the corresponding quantile of their own preference-conflict distribution.

135 HIR-Bench establishes constructively that within-population structure can become consequential for
 136 candidate choice. The failed transfer sets its limit: the corresponding regime in biological data cannot be
 137 inferred from this synthetic boundary.

138 **Supplementary Note 3. Robustness of the natural-heterogeneity and large-scale analyses**

139 The main text uses patient-derived glioblastoma and Tahoe-100M to ask whether the information re-
 140 quirements identified in the controlled experiments are also present when cellular heterogeneity is not
 141 created by combining experimentally distinct source populations. These datasets measure differential
 142 response, recoverability and the extent to which state-specific response ordering leaves headroom for
 143 population structure to alter candidate preference; no retrieval is benchmarked on them.

144 **Response alignment between compartments.** For 17 patient–drug pairs in which both compartments
 145 carried enough treated and control cells, the cosine between the malignant and myeloid perturbation-
 146 response directions has a median of 0.566, against a range of 0.02 to 0.39 for the constructed mixtures
 147 used elsewhere in this study. Read directly, that contrast says the two compartments of a real tumour
 148 respond far more alike than two experimentally combined cell lines do.

149 The cosine is descriptive. It is confounded with effect size and with signal-to-noise: a compartment
 150 with a weak or noisily estimated response yields a lower cosine for reasons that have nothing to do with

its pharmacology, and the compartments here differ substantially in cell number. The two natural-tissue results the main text relies on, the recoverability gap and the conservation of candidate ordering across disjoint compartments, are measured without reference to it.

Robustness of glioblastoma compartment assignment. The ZhaoSims2021 dataset does not provide resolved cell-type labels for the analysed cells, so malignant, tumour-associated myeloid and oligodendrocyte compartments were assigned from marker scores. Cells with fewer than 200 detected genes were removed, counts were library-size normalized to 10^4 and $\log(1 + x)$ transformed, and gene symbols were upper-cased; compartment scoring runs on that full matrix, before the 2,000-gene selection used downstream, so the labels do not depend on a gene set chosen for downstream variance. Three marker sets were used: malignant *SOX2*, *EGFR*, *PDGFRA*, *OLIG1*, *OLIG2*, *GFAP* and *AQP4*; myeloid *PTPRC*, *CD14*, *AIF1*, *CSF1R*, *ITGAM*, *C1QA* and *C1QB*; oligodendrocyte *MBP*, *PLP1*, *MOG* and *MAG*. A marker absent from the matrix is skipped. Each gene was z-scored across all cells of the cohort and the compartment score is the mean of its set’s z-scores, so the scores are cohort-dependent by construction. A cell was called for the highest-scoring compartment only if the mean log-normalized expression of that compartment’s key markers (malignant *SOX2*, *OLIG2*, *EGFR*; myeloid *CD14*, *CSF1R*, *C1QB*, *AIF1*; oligodendrocyte *MBP*, *PLP1*) reached 0.25 and its z-score margin over the second-highest compartment reached 0.25. Cells failing either criterion were left unassigned rather than swept into a leftover class, which leaves 39.9% of cells unassigned. The build refuses to proceed unless every called compartment expresses its own key markers most strongly, and writes the check to `results/zhao_gbm/compartment_validation.csv`.

The results were stable when both thresholds were varied jointly over

$$\{0.10, 0.25, 0.40, 0.50\}.$$

The unsupervised arm is the best of four partitions of the same cells: k -means with $k = 2$, a two-component Gaussian mixture, Leiden¹ and HDBSCAN². Leiden is the maximum on 152 of the 180 natural splits, so the reported best-unsupervised value is close to Leiden’s alone.

Across this range, the excluded fraction varied from 29% to 58%, while the supervised drug-response ceiling remained between 0.915 and 0.923, the best unsupervised recovery between 0.77 and 0.80, and the paired recoverability gap between approximately 0.11 and 0.13. The median cosine between malignant- and myeloid-specific perturbation-response directions remained between 0.54 and 0.59. Thus, the natural-tissue conclusions are not specific to the primary 0.25 assignment threshold.

Patient-level uncertainty. The natural-tissue recoverability analysis contains 36 within-patient drug-pair splits across four patients and two compartments, with 30 of the 36 splits contributed by a single patient. Drug-pair splits were therefore not treated as independent replicates. Patient-level cluster bootstrap resampling was used to quantify uncertainty, and the analysis was repeated after removing the dominant patient.

The reported recoverability gap is computed within each split before aggregation,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

rather than as the difference between the two marginal medians. The primary paired median is +0.117; subtracting the marginal medians would instead give +0.147. Across 5,000 patient-cluster bootstrap resamples, the natural-minus-constructed difference in recoverability gap was +0.109 with an interval of $[+0.052, +0.129]$, and no bootstrap draw was at or below zero. Because only four patient clusters contribute to this calculation, the interval is interpreted as supportive rather than as a precise population-level uncertainty estimate.

Two aggregation controls give the same gap. Fixing every probe–clusterer pair, so that the same pair is scored in both arms, gives +0.122 in tissue against -0.002 in the constructed arm. Removing the patient that contributes 30 of the 36 splits gives +0.123, with a $[+0.098, +0.138]$ interval.

Large-scale measurement in Tahoe-100M. Tahoe-100M plate 3 provides a substantially larger setting in which heterogeneity occurs within cell lines rather than through constructed mixtures. Two cellular-state definitions are used below, and both are fixed before any treated cell is examined. The first is the released cell-cycle phase call, G1 against G2M, with S-phase cells dropped as intermediate and each state’s response referred to its own phase-matched control. The second is k -means with $k = 2$ fitted on each cell line’s DMSO_TF control cells in the 2,304 highly variable genes in log1p space, with ten initializations and a fixed seed; treated cells are assigned to the nearest control centroid, the larger control cluster is taken as the first state, and a cell line enters only when it holds at least 100 control cells. Across 3,630 evaluable drug–cell-line conditions, the median cosine between the two state-specific response directions was 0.739 (interquartile range 0.600–0.836). Fewer than 0.11% of conditions were as divergent as the constructed mixtures used in the controlled experiments.

This estimate is not driven materially by the minimum-cell filter. Conditions were excluded when one of the required state-specific treated or control arms contained fewer than 50 cells, which can preferentially remove strong cell-cycle-arresting perturbations. Across drugs, coverage was only weakly associated with induced-response cosine (Spearman $\rho = 0.077$), and restricting the analysis to well-covered compounds changed the median from 0.739 to 0.740.

Recoverability remained measurable at scale. Across 960 sampled drug pairs and 48 cell lines, the supervised ceiling was 0.853 and the best unsupervised procedure reached 0.611, giving a median paired gap of 0.157. The gap was positive in all 48 evaluable cell lines. Individual unsupervised methods gave similar results: k -means 0.592, Gaussian mixtures 0.582 and Leiden¹ 0.563.

State-specific ordering and decision-relevance headroom. We finally compared relative drug-response ordering between non-overlapping cellular states. Under the cell-cycle partition, the median disjoint Spearman correlation between the two drug–drug similarity structures was 0.841 across 44 evaluable contexts. Under the independently control-derived two-state partition, the corresponding median was 0.778 across 45 contexts.

These high correlations indicate that relative drug ordering is often substantially preserved between cellular states even when the states respond differently. The pattern is not universal: under the control-derived partition, 2 of 45 contexts had a correlation below 0.5, with a minimum of 0.236. Such contexts leave greater headroom for state-specific response information to alter candidate ordering. Headroom of that kind is what a population-level method would need in order to change the selected candidate, and it is measured here on the ordering alone.

The two state definitions agree only moderately in their ranking of contexts (Spearman $\rho = 0.466$), reinforcing that apparent decision-relevance headroom depends on how cellular states are defined. Both partitions were specified independently of downstream retrieval performance.

Tahoe-100M therefore supports the same separation of concepts observed elsewhere in the study: state-specific responses can exist, and response identity can remain recoverable, while much of the relative candidate ordering is still preserved across cellular states. These are intrinsic-state diagnostics: they constrain the conditions under which population-level retrieval could add value. The advantage itself is measured on the same plate by the source-to-target intervention benchmark of the main text (Figs. 5 and 6).

Supplementary Note 4. Perturbation-predictor diagnostics

The predictor analyses ask one question: whether a predicted response population contains state-specific perturbation effects that a downstream population-aware retriever could exploit.

Additive response models. For a predictor that applies the same perturbation effect δ_d to every cell in a context,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

the response of any two cellular states is identical after subtraction of their corresponding controls,

$$r_{d,0} = r_{d,1} = \delta_d.$$

241 Consequently,

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

242 This result is independent of the heterogeneity of the starting control population. An additive predictor
243 can therefore generate a visually heterogeneous predicted population while containing no state-specific
244 variation in the perturbation effect itself.

245 The average-effect and linear-latent predictors used in the main analysis satisfy this additive limit
246 by construction when the perturbation response is referenced to the model-consistent baseline. They are
247 included to establish this limiting case.

248 **The predict-then-rank construction.** The predictor-generated retrieval used in this note factorizes the
249 pipeline the field actually uses into a forward predictor and a retrieval rule, so that the retrieval layer
250 can be varied with the predictor held fixed. It is run on SciPlex3 and is not the Tahoe-100M prediction
251 arm of the main text. Queries are cross-line: for each of the three SciPlex3 cell-line pairs (K562–A549,
252 A549–MCF7, K562–MCF7) and each of ten drugs selected by a cross-context divergence probe, the query
253 is that drug’s response blended across the two lines at mixing fraction $\alpha \in \{0.5, 0.7, 0.9\}$ over 200 cells,
254 with eight seeds, giving 720 queries. The candidate library is the query’s own drug together with up to
255 30 further drugs shared by both lines; the query’s drug is the single relevant candidate. Each predictor is
256 asked to imagine every candidate’s response in the query’s real blended control cells, mixed at the same
257 α , so no predictor sees the within-query line structure. The imagined populations are then ranked by
258 four retrieval rules: mean cosine, R^2 of the predicted against the observed mean delta, energy distance
259 and coverage. Performance is nDCG@10 with binary relevance, and the reported quantity is the distri-
260 butional minus the mean-cosine value for the same predictor, so each predictor is its own control. The
261 predictors are fitted on the full SciPlex3 matrix, including the lines that compose each query, so these
262 values are an in-sample ceiling on candidate generation rather than a transfer result.

263 **Two reference baselines.** A predicted population’s induced response can be referred to two different
264 controls, and the two answer different questions. Referring it to the model’s own zero-effect baseline,
265 which is its reconstruction of the control population whenever the model decodes and the control cells
266 themselves when it does not, cancels the reconstruction bias and leaves the divergence the model itself
267 generates. The additivity result above pins this quantity to exactly 1 for any predictor of the form $\hat{x}_{i,d} =$
268 $g(x_i) + \delta_d$ with a fixed g , covering both the gene-space case $g = \text{id}$ and the latent case in which g is the
269 model’s own reconstruction, so the value is analytic and not estimated. Referring it to the real control
270 cells is instead what a retrieval scorer does, and the resulting cosine contains the model’s reconstruction
271 bias, which differs between cellular states and is not a perturbation response. Across the same 30 cell-
272 line and drug combinations, the average-effect, linear-latent, scGen and optimal-transport predictors
273 give 1.000, 1.000, 0.972 and 0.900 against the model baseline, and 1.000, 0.190, 0.404 and 0.380 against the
274 real control; real treated cells give 0.205, and have no model baseline.

275 The two columns are informative together. Lower cosine means more state-specific divergence. The
276 average-effect predictor adds its effect directly in gene space and has no reconstruction step, so its two
277 references coincide at 1.000, which is the correct answer for such a model. The other three pass their
278 output through a decoder whose reconstruction error differs between cellular states, and that error enters
279 the real-control cosine as if it were a perturbation effect. The linear-latent model is the clearest case: it
280 generates exactly zero state-specific divergence, yet measured the way a scorer measures it it reads 0.190,
281 marginally more divergent than the 0.205 of the real cells it replaces. A distributional score run on such
282 a predicted population is therefore partly scoring the decoder. This is a second reason, independent of
283 transfer accuracy, why predicted candidates do not provide a clean test of whether population structure
284 improves retrieval.

285 **scGen.** A deployed scGen model was evaluated using the same within-context cellular-state partition
286 and state-specific induced-response definition as the other predictors. Across the same 30 cell-line and
287 drug combinations, its mean induced-response cosine against its own reconstruction baseline was 0.972,
288 close to the additive limit of 1.000, and 0.404 against the real control. Thus, in this experiment scGen

289 preserved substantially less state dependence in the perturbation effect than the 0.205 observed in real
290 treated cells, under either reference.

291 The model is the published `scgen`.SCGEN implementation on `scvi-tools`³, trained on the SciPlex3
292 tensor with the perturbation label as batch key and the cell line as label key, 30 latent dimensions, batch
293 size 512, at most 40 epochs with early stopping after 15 epochs without improvement, and a fixed seed.
294 Predictions use `scGen`'s own latent arithmetic: one latent delta per perturbation, taken as the mean latent
295 of treated cells minus the mean latent of control cells, added to the encoded control mean and decoded.

296 **Entropic optimal-transport mapping.** We additionally evaluated an entropic optimal-transport map as
297 a predictor that is not constrained to apply a shared additive shift to all cells. It is a different map from
298 the Gaussian (Bures) predictor of the Tahoe-100M arm (Methods). Under the within-context differential-
299 response analysis, the optimal-transport predictions reached an induced-response cosine of approxi-
300 mately 0.900 ± 0.107 against their own reconstruction baseline, moving away from the additive limit but
301 remaining much more aligned than the observed responses.

302 A latent-space implementation passed the corresponding perturbation-learning diagnostic (mean-
303 delta agreement 0.352), whereas the same construction applied directly in gene space showed substan-
304 tially weaker perturbation learning (0.057). The latent-space implementation was therefore used for the
305 differential-response comparison.

306 The map is an entropic (Sinkhorn) coupling⁴ between control and treated cells, solved with POT⁵,
307 with uniform marginals, regularization 0.05, at most 500 iterations and a cost matrix rescaled by its
308 maximum, solved in a 30-component PCA latent fitted unsupervised on at most 20,000 cells. Each fitting
309 context is centred on its own control mean, and the query context is excluded from the fit. A barycentric
310 projection gives a per-source-cell displacement, extended to the query's real control cells by nearest-
311 source matching in the same centred space and decoded through the PCA inverse transform.

312 Opening this state-specific response channel was not sufficient to provide a clean retrieval test. When
313 the same entropic map was used to generate candidates in the cross-context predict-then-rank setting,
314 Hit@1 was 0.053 under mean cosine and 0.007 and 0.008 under the two distributional rules, against
315 0.879 to 0.899 for the additive average-effect candidate source. The optimal-transport predictions were
316 therefore too inaccurate in that transfer setting to test whether greater state dependence itself improves
317 retrieval.

318 Taken together, these analyses leave a specific unresolved experiment. A decisive test requires a
319 predictor that both generalizes accurately to the target cellular context and reproduces the state-specific
320 perturbation responses present in real cells. No predictor evaluated here satisfied both requirements
321 simultaneously.

Supplementary Table 1. Retrieval scores, task constructions and evaluation criteria.

(a) Retrieval scores.

Score	Input representation	Ranking rule	Role
Mean cosine	Control-referenced mean perturbation signatures	Descending cosine similarity between query and candidate mean-response vectors	Primary mean-signature baseline
CMap-style cosine	Same control-referenced mean differential-expression vectors as mean cosine	Same cosine operation as the implemented mean-cosine baseline	Signature-retrieval reference; numerically identical to mean cosine in this implementation
WTCS	Mean-collapsed differential-expression response	Rank-based weighted connectivity / enrichment score	Canonical rank-based signature baseline; related to but not algebraically identical to mean cosine
Energy distance	Empirical control-referenced single-cell response populations	Candidates ranked by increasing energy distance to the query population	Primary population-level retrieval score
RBF-MMD	Empirical response populations	Candidates ranked by increasing maximum mean discrepancy under an RBF kernel	Alternative distribution-sensitive score

Supplementary Table 1(a) continued.

Score	Input representation	Ranking rule	Role
Sliced Wasserstein	Empirical response populations	Candidates ranked by increasing sliced Wasserstein distance using 128 random projections	Alternative distribution-sensitive score
Subpopulation coverage	Response populations decomposed into response components	Component-level distances aggregated with temperature β ; increasing β gives progressively greater weight to the least well-matched response component	Population-sensitive score used for the partial-observation response-matching analyses

322 The four score families become five population-level configurations because subpopulation-coverage contributes two:
323 coverage-mean is the $\beta \rightarrow 0$ limit of the aggregate and coverage-worst the $\beta \rightarrow \infty$ limit, both with energy distance as the
324 per-component base metric. The CMap-style cosine is numerically identical to mean cosine on control-referenced mean
325 vectors; its sign-flipped variant is scored in the partial-observation experiments as an anti-correlation floor and is not reported
326 here.

Supplementary Table 1 continued. (b) Retrieval task constructions.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Observed-population retrieval over heterogeneous queries	A heterogeneous query population combines two response modes at weight α ; the library holds a held-out re-mixture at the same weight, the two single-mode populations, and 40 distractors, giving a 43-candidate library	Hit@1; only the both-mode candidate is relevant, so a single-mode population counts as a miss even when it carries one of the query's two response modes	Retrieval query; headline values macro-averaged over seven task-setting combinations
Controlled heterogeneous queries	Two response modes are combined at mixing weight α ; controlled SciPlex3 analyses use HDAC- and JAK-associated response sources	Source identification and response-matching criteria	Query / task setting
Partial-observation retrieval	Only the response information available under the controlled partial-observation protocol is exposed to the retriever; population and mean methods rank the identical candidate library once	The resulting fixed ranking is evaluated under response matching, MoA recovery and minority-state coverage	765 total queries
Mechanism-evaluable subset	Same partial-observation rankings; restricted to queries for which the query mechanism remains represented among candidates	MoA-nDCG	600 queries; 165 excluded because MoA recovery is undefined
Real-data minority-state projection	Top-ranked population and mean candidates evaluated on 239 task instances spanning constructed cross-line, within-line, Frangieh, CD34+ and predicted-population settings	Difference in minority-state coverage	Task-level difference; not interpreted as 239 independent biological replicates
GDSC2-linked retrieval	Held-out SciPlex3 compound at 10 μ M used as query; remaining matched compounds form a 34–35 candidate library	Spearman agreement between RNA-derived ranking and external GDSC2 functional-similarity ordering	Per-query correlation summarized within SciPlex3 cell line

Supplementary Table 1 continued. (c) Evaluation criteria applied to fixed rankings.

Criterion	Definition	Interpretation
Hit@1	Indicator that the both-mode candidate, the held-out re-mixture at the query's own weight, is ranked first	Recovery of the query's response composition, not of its perturbation identity alone
Response-matching regret	Energy-based welfare regret against the latent per-cell objective used to construct the controlled partial-observation task; reported as regret reduction relative to mean retrieval	Response fidelity; intentionally close to the population information used by the retriever
MoA-nDCG	Normalized discounted cumulative gain rewarding candidates that share the query perturbation's annotated mechanism	Task-proximal biological criterion; undefined when the relevant mechanism is absent from the candidate library
Minority-state coverage	Measures whether the selected candidate better preserves the response of the less-represented cellular state	Task-proximal measure of coverage of heterogeneous response structure
GDSC2 functional similarity	Spearman correlation between cell-line-centred cross-cell-line AUC profiles of the query and candidate drugs	Externally measured functional similarity; not absolute potency
Protein mean evaluator	Cosine similarity between control-referenced mean protein-response vectors	External evaluator preserving only mean protein response
Protein population evaluator	Negative energy distance between control-referenced per-cell protein-response populations	External evaluator retaining cell-to-cell protein-response structure
Response-magnitude matching	$-\ \bar{d}_q\ - \ \bar{d}_c\ $	Query-dependent scalar control that uses no within-population information

Query states s are obtained by k -means with $k = 2$ on the query cells, the minority state being the smaller cluster. A candidate d with response population P_d receives the welfare $U(d) = \text{agg}_s(-\mathcal{E}(P_d, Q_s))$, where $\mathcal{E}$ is the energy distance and the aggregator is the minimum over states in the worst-case form and the mean in the mean form. The response-matching regret of a selected candidate $\hat{d}$ is $\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0$, taken over the observed library $\mathcal{L}$, so the reference is the best available candidate rather than an unobservable optimum. Minority-state coverage is $\frac{1}{2}(1 + \cos(\bar{P}_d, Q_{\min}))$, the cosine between the candidate's mean response and the mean response of the query's minority state rescaled to $[0, 1]$. The same definitions are given in the main Methods.

Supplementary Table 2. Data resources, preprocessing and analytical roles.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
SciPlex3	276,325 cells; 2,000 genes	188 compounds shared across A549, K562 and MCF7	Log-transformed 2,000-gene representation; perturbation responses referenced to matched controls	Primary observed-population drug-retrieval setting; controlled within- and cross-context constructions; source of the GDSC2-linked transcriptomic rankings
Frangieh Perturb-CITE-seq	218,023 RNA-profiled cells for retrieval and 218,331 barcode-matched cells for the RNA-protein evaluation; 2,000 genes; 20 retained surface proteins	239 CRISPR knock-outs across Control, IFN γ and Co-culture conditions	RNA and protein modalities joined by cell barcode; responses referenced to condition-matched controls	Independent genetic-perturbation retrieval setting and matched RNA-protein evaluator-shape experiment; not treated as a drug-retrieval benchmark
CD34+ HSPC	33,984 cells; 2,000 genes	36 compounds	Log-transformed expression; DMSO-referenced perturbation responses	Negative-control retrieval setting for testing whether population-level information consistently improves ranking

Supplementary Table 2 continued.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
ZhaoSims2021 glioblastoma	10 patients; 6 compounds	Acute-slice culture and surgical biopsy with co-existing malignant, myeloid and oligodendrocyte compartments	Counts normalized to 10^4 per cell, log1p-transformed and reduced to 2,000 highly variable genes; state-specific responses referenced within patient and compartment	Natural-tissue analysis of differential response, recoverability and state-specific response ordering; no retrieval benchmark is constructed
Tahoe-100M plate 3	4,158,278 cells; 2,304 genes	50 cell lines $\times$ 93 compounds at one concentration, with DMSO_TF controls	Released 2,304-gene log1p representation used without refitting; state-specific responses referenced to matched state controls	Two roles: large-scale measurement of differential response, recoverability and state-ordering agreement under intrinsic within-cell-line heterogeneity, and the substrate for the source-to-target intervention retrieval benchmark (Figs. 5 and 6)
GDSC2	286 compounds across 969 cell lines in the retained release	35 compounds matched to the SciPlex3 panel; A549, K562 and MCF7 excluded from external profiles	Fitted dose-response AUC; primary functional similarity uses cell-line-centred cross-line response profiles	External functional evaluation of fixed SciPlex3 RNA-derived rankings
HIR-Bench	13,440 synthetic instances	672 parameter configurations $\times$ 20 seeds in a two-subpopulation latent-welfare model	Synthetic observed, predicted-mean and predicted-structure information conditions	Controlled demonstration that population structure can alter candidate preference; not interpreted as a biological generative model

The resources above answer distinct questions and are not pooled as independent replicates of a single benchmark. Dataset-specific cell-number thresholds, subsampling rules and statistical units are given in the corresponding Methods sections.

Supplementary Table 3. Sensitivity of the GDSC2 functional-similarity evaluation to response-profile construction.

The primary external functional criterion compares drug–drug response profiles after removing the shared cell-line sensitivity effect. For cell line l and drug d ,

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in that cell line. Drug–drug functional similarity is then the Spearman correlation between centred cross-cell-line response profiles. The table compares this primary construction with the corresponding uncentred AUC-profile similarity.

RNA-derived ranking rule	Cell-line-centred functional similarity	Uncentred functional similarity
Energy distance	+0.265	+0.0902
Mean cosine, control referenced	+0.083	+0.138
Mean cosine, treated mean only	+0.241	+0.092
Response-magnitude matching	+0.232	+0.217
Candidate magnitude only	−0.330	+0.135
Potency matching diagnostic	+0.399	+0.290
Energy, magnitude-adjusted partial rank correlation	+0.105	n/a

Values are median per-query Spearman correlations over 103 leave-one-drug-out queries across the three SciPlex3 cell lines. The primary centred functional criterion is constructed from GDSC2 AUC profiles after subtracting each cell line’s mean response

across all 286 compounds in the retained release. Drug pairs sharing fewer than 100 cell lines are excluded; 591 usable drug pairs remain, with a median of 909 shared cell lines per pair.

The change between centred and uncentred constructions shows that external evaluation is itself sensitive to how the functional measurement is summarized. Under the centred criterion, energy retrieval exceeds the control-referenced mean-signature baseline, but much of this association is also reproduced by response-magnitude matching. Under the uncentred criterion, the ordering between energy and the mean-signature baseline reverses. Neither construction is interpreted as a direct measure of therapeutic utility.

Supplementary Table 4. Perturbation-predictor implementations and their role in the differential-response analysis.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Average-effect	Per-drug mean perturbation delta	Predicted delta applied to the real control cells of the query context	Predict-then-rank candidate source and within-context differential-response reference	Additive by construction; state-specific induced-response cosine equals 1
Nearest-neighbour	Transfers the perturbation effect from the training context with the most similar untreated control state	Transferred effect applied to real query-context control cells	Predict-then-rank candidate source	Tests simple context transfer rather than a general virtual-cell architecture
Linear-latent	Additive latent arithmetic in an encoded representation followed by linear decoding	Observed control cells encoded, shifted in latent space and decoded individually	Predict-then-rank candidate source and additive latent reference	Can preserve baseline cellular heterogeneity while remaining additive in perturbation effect
scGen	Variational-autoencoder perturbation model with latent arithmetic	Evaluated on the same within-context state partition and state-specific response definition used for other predictors	Additional differential-response diagnostic reported in the main text	Mean induced-response cosine 0.972 over 30 cell-line and drug combinations
Entropic optimal-transport map	Non-parametric Sinkhorn mapping that permits cell-dependent displacements	Control cells mapped individually to predicted treated states	Additional non-additive differential-response diagnostic; also tested as a candidate-response source	Within-context induced-response cosine approximately 0.900 ± 0.107 against the map's own reconstruction baseline and 0.380 against the real control; cross-context candidate predictions were too inaccurate for a clean retrieval test

Supplementary Table 4 continued. Implementation of each predictor.

Predictor	Implementation
Average-effect	Per drug d , $\hat{\delta}_d = \text{mean}_c (\text{mean}(\text{treated}_{c,d}) - \text{mean}(\text{control}_c))$ over the SciPlex3 cell lines c , a line contributing only where it has at least two treated cells for d ; one signature per drug, independent of the query context. The predicted population is that single signature added to each of the query context's real control cells, so every cell is shifted identically. No reconstruction step, so the model's own zero-effect baseline is the control cells themselves. A drug absent from the fit is scored at -10^6 under every rule rather than imputed
Nearest-neighbour	$c^* = \arg \min_c (1 - \cos(\bar{x}_{\text{control}}^{\text{query}}, \bar{x}_{\text{control}}^c))$ over the training lines, then $\hat{\delta}_d = \text{mean}(\text{treated}_{c^*,d}) - \text{mean}(\text{control}_{c^*})$. Distances are compared strictly, so an exact tie is resolved by context iteration order; where d was not measured in c^* the signature falls back to the mean of d 's signatures over the remaining admissible lines. Population synthesis as for average-effect. The two query-composing cell lines remained eligible donor contexts, so this experiment is interpreted as an in-sample ceiling rather than as a transfer evaluation

Supplementary Table 4 continued.

Predictor	Implementation
Linear-latent	PCA with 30 components fitted, unsupervised and once, on up to 20,000 cells drawn from the full SciPlex3 matrix at seed 0. Each line's control latent mean is stored; each drug's latent delta is the mean over lines of (mean latent of treated – that line's control latent mean), again requiring at least two treated cells. Prediction encodes each real control cell of the query context, adds the drug's latent delta to that cell's own latent code, and decodes each cell by the PCA inverse transform. The decoder is linear, so the composite map is $\text{dec}(\text{enc}(x) + dz) = Px + Pdz$ and the predictor is additive in gene space and deterministic. Because it reconstructs, its own zero-effect baseline is the PCA reconstruction of the control cells rather than those cells
scGen	Published <code>scgen.SCGEN</code> on <code>scvi-tools</code> ; trained on the SciPlex3 tensor, batch key = perturbation, label key = cell line; 30 latent dimensions; batch size 512; at most 40 epochs with early stopping after 15 epochs without improvement; fixed seed; per-perturbation latent delta added to the encoded control mean and decoded
Optimal transport	Entropic (Sinkhorn) coupling in a 30-component PCA latent; cost rescaled by its maximum; uniform marginals; regularization 0.05; at most 500 iterations; fitted per non-query context centred on that context's own control mean, with the query context excluded; barycentric projection extended to the query's control cells by nearest-source matching and decoded by the PCA inverse transform

345 Predictors are used as candidate-response sources or information diagnostics rather than as a benchmark of perturbation-
 346 prediction architectures. The reported predict-then-rank experiment uses predictors fitted on the full SciPlex3 matrix and is
 347 therefore interpreted as an in-sample ceiling, not as a cross-context generalization benchmark.
 348 Constants shared by every row of the predict-then-rank run: 720 cross-line queries, formed from the three cell-line pairs (K562–
 349 A549, A549–MCF7, K562–MCF7) at mixing weights $\alpha \in \{0.5, 0.7, 0.9\}$ over 10 divergent drugs and 8 seeds; 200 query cells
 350 per query; a 31-drug candidate library holding the true drug and 30 other shared drugs; 200 predicted cells synthesized per
 351 candidate; query construction seeded at $1000 + s$ and distractor selection at $7000 + s$ for $s = 0, \dots, 7$. The within-context
 352 differential-response measurements in Supplementary Note 4 are a separate run over 30 cell-line and drug combinations with
 353 400 control cells per context split into two states, a state requiring 60 cells and a treated arm 40, at seed 0. Nearest-neighbour
 354 appears only in the predict-then-rank run and scGen only in the differential-response run; the entropic optimal-transport map
 355 appears in both.

356 **Supplementary References**

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